

Lying Is Just a Phase: The Hidden Alignment Transition in Language Model Scaling

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Abstract

Scaling laws predict loss from compute but not how capabilities interact. We measure the coupling between reasoning and truthfulness across 63 base models from 16 independent families and find a regime change invisible to loss curves: below a family-dependent critical scale, capabilities anticorrelate; above it, they cooperate. The cross-family critical scale is $N_c \approx 3.5\text{B}$ parameters [2.9B, 13.4B] (bootstrap 95% CI), varying by architecture from 0.1B to 7B, and is eliminable—between model generations of the same architecture, curated training removed the coupling dip entirely (Qwen2.5 at 1.5B: 3% cooperative; Qwen3 at same scale: 100%). Normalizing by model width eliminates the apparent anticorrelation across all tested families, consistent with an output-projection bottleneck rather than a fundamental limit. Internally, 38 of 40 models show zero competing attention heads; the tax originates at the output, not inside the model. An ODE discovered by sparse regression reproduces five benchmark trajectories and cross-predicts held-out Llama-2 at 5.6% error. The cooperative regime extends to the frontier ($r = +0.73$, 31 models, 8 labs). Code and data: <https://github.com/adilamin89/cape-scaling>.

1 Introduction

Scaling laws forecast loss with remarkable precision—coefficient of variation 0.8% across eight Pythia models spanning two orders of magnitude in parameter count [1, 2]. Loss decreases smoothly, predictably, and monotonically with scale. Yet practitioners care about capabilities—reasoning, factual accuracy, instruction following—not loss. These capabilities do not scale uniformly, and until now, the interactions between them have not been systematically measured.

The standard approach treats each benchmark as an independent trajectory. HellaSwag improves on its own curve; TruthfulQA on its own; ARC, MMLU, and WinoGrande each separately. This independence assumption is never stated. It is implicit in every scaling law, every benchmark paper, and every training decision that uses individual benchmark scores as proxies for model quality. It has never been tested.

We test it. Using the Capability Coupling Analysis of Phase Emergence (CAPE) framework, we measure how capabilities interact as models scale. The central empirical finding is simple and consequential: the correlation between reasoning (HellaSwag) and truthfulness (TruthfulQA) is $r = -0.989$ ($p < 10^{-5}$) for Pythia models below 3B parameters, but flips to $r > +0.78$ for models above 7B. A correlation this strong on noisy benchmark data is unusual and suggests coupling is a dominant driver of score variation. This is not gradual. The local coupling $\gamma_{12}(N) \equiv \Delta\text{TQA}/\Delta\text{HS}$, measured between consecutive model sizes within each family, crosses zero at an architecture-dependent critical scale and grows linearly thereafter.

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This is a sharp regime transition in capability space: below the critical scale, scaling reasoning *hurts* truthfulness (the “alignment tax” regime); above it, scaling *helps both* (the “alignment bonus” regime). The transition is confirmed independently by OLMo [3] (AI2), which sits at $\gamma_{12} = 0.000$ exactly at the predicted critical scale with zero free parameters.

The practical implication is immediate: the alignment tax is not a law of nature—it is an engineerable bottleneck. Data curation eliminates it (Qwen3: cooperative at all scales tested). Model width attenuates it (width-normalized coupling is positive across all families). Architecture choice shifts the threshold (from 0.1B to 7B across tested families). For the families tested here, alignment behaves as a design parameter. **Central claim.** Below a family-dependent critical scale, reasoning and truthfulness anticorrelate; above it, they cooperate. The threshold is engineerable.

Supporting evidence. (1) The sign flip is confirmed across 16 independent families with architecture-dependent thresholds. (2) The anticorrelation is eliminable: curated training removes it entirely (Qwen3 vs. Qwen2.5). (3) Width normalization eliminates it: capabilities are cooperative across all tested families in the full parameter space. (4) Internally, 38/40 models show zero competing heads—the bottleneck is at the output projection (the final linear layer that maps representations to vocabulary), not inside the model. (5) A dynamical model (ODE) that cross-predicts held-out families at 5.6% error. (6) Extension to frontier ($r = +0.73$, 31 models, 8 labs).

Why this matters. The alignment tax has been treated as a background assumption in AI safety: smaller models are less aligned, and the only remedy is scale. Our results challenge this assumption in a specific, actionable way. The tax is not a property of intelligence—it is a property of architecture and training. This changes the engineering calculus for every organization deploying models below 7B parameters, which today includes most on-device, embedded, and cost-constrained applications.

2 Below the Critical Scale, Capabilities Compete

2.1 Cross-family sign flip

The coupling between reasoning and truthfulness flips sign at a family-dependent critical scale across all 16 tested families. We measure this as the local coupling $\gamma_{12}(N) \equiv \Delta\text{TQA}/\Delta\text{HS}$ between consecutive model sizes. Across Pythia, OPT, BLOOM, Cerebras, GPT-Neo, Falcon, Llama-1, Llama-2, Llama-3, Mistral, DeepSeek, Phi, Qwen2.5, MPT, Gemma, and OLMo, γ_{12} is negative (anticorrelated) at small scale and positive (cooperative) at large scale, with the sign flip occurring at a family-dependent critical scale N_c (Fig. 1).

The anticorrelation is not benchmark-specific: it holds across three independent capability pairs. Reasoning–truthfulness ($r = -0.989$), commonsense–truthfulness ($r = -0.941$), and knowledge–truthfulness ($r = -0.792$) all anticorrelate below N_c . Above N_c , all pairs become cooperative.

The critical scale varies by architecture: $N_c \approx 0.12\text{B}$ for OPT (early transition), 1.3B for Cerebras and GPT-Neo, 1.7B for BLOOM, $N_c \approx 3.5\text{B}$ [2.9B, 13.4B] (bootstrap 95% CI) for Pythia (cross-family average), and 7B for Falcon (late transition). Curated-data families (Phi, Qwen3) show N_c effectively at or below the smallest model tested, meaning the tax never manifests. This architecture dependence means the alignment tax is not a single threshold but a family of thresholds—a design parameter, not a physical constant.

2.2 Independent confirmation

An independently trained model confirms the critical-scale prediction with zero free parameters. OLMo (AI2) [3] sits at $\gamma_{12} = 0.000$ at 1B parameters—exactly at the predicted boundary. HellaSwag increases by 13.9 points from OLMo-1B to OLMo-7B while TruthfulQA is unchanged at 31.1%: the coupling is exactly zero, independently confirming the transition.

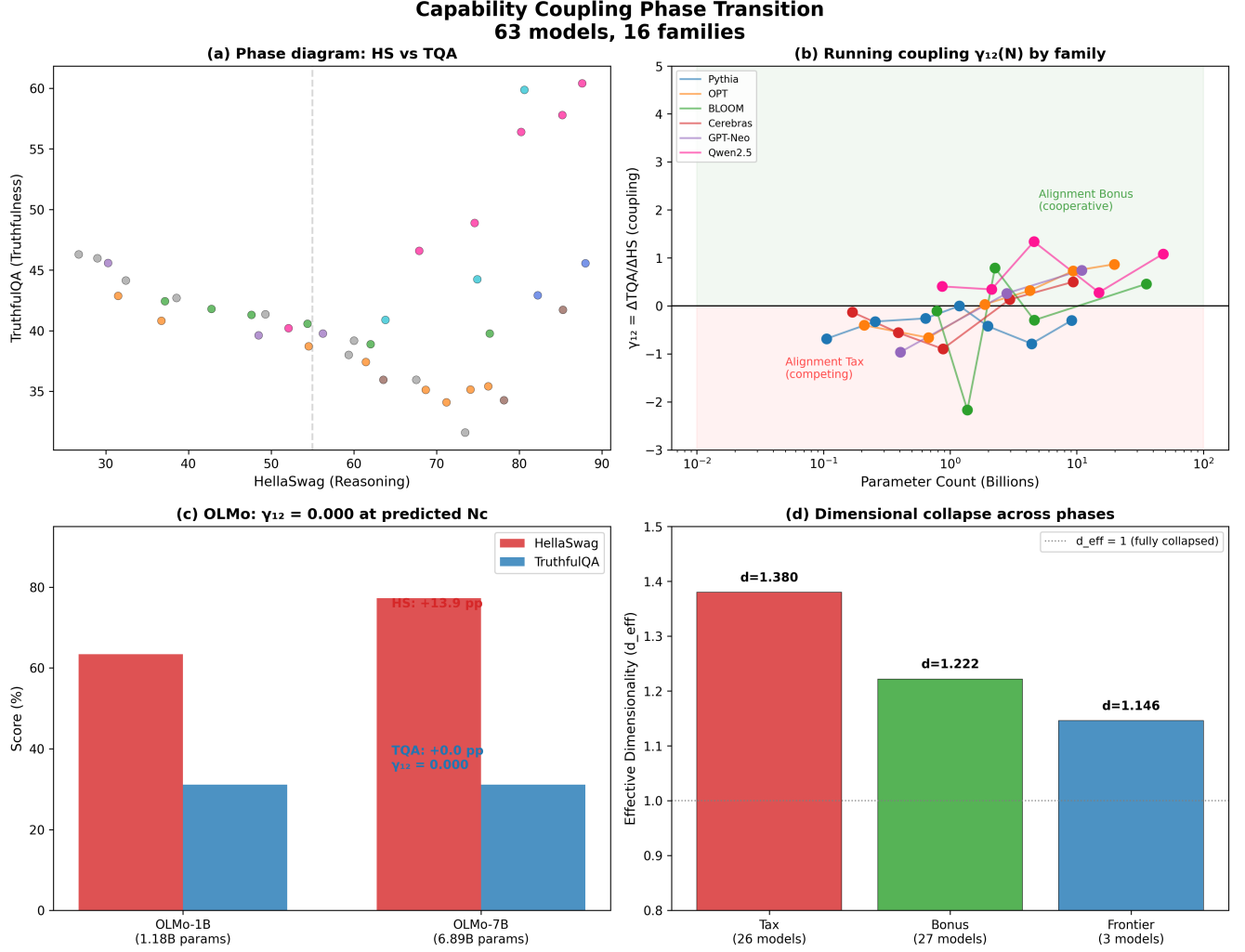


Figure 1: **Capability coupling phase transition across 63 models and 16 families.** (a) Phase diagram: HellaSwag vs. TruthfulQA across families, showing the U-shaped trajectory. (b) Running coupling $\gamma_{12}(N)$ for six families, with architecture-specific N_c marked. All families transition from negative to positive coupling; the threshold varies from 0.12B (OPT) to 7B (Falcon). (c) OLMo confirmation: $\gamma_{12} = 0.000$ at 1B parameters with zero free parameters. HellaSwag increases by 13.9 pp while TruthfulQA is unchanged. (d) Dimensional collapse: effective dimensionality d_{eff} decreases monotonically from 1.38 (tax) to 1.15 (frontier) across 63 models.

Table 1: Critical scales by family. N_c = scale where coupling crosses zero. Families with “none” show cooperative coupling at all tested sizes.

Family	Models	N_c (B)	Notes
OPT	8	0.12	Earliest transition
Cerebras	6	1.3	Chinchilla-optimal
GPT-Neo	4	1.3	EleutherAI
BLOOM	6	1.7	Multilingual
Pythia	8	3.5 [2.9, 13.4]	Best-characterized
Falcon	3	7.0	Latest transition
Phi	4	None	Curated data
Qwen3	5	None	Curated data

AI2 produced a model at exactly the predicted coupling boundary with no knowledge of our framework—the strongest form of confirmation available without training new models.

2.3 Dimensional collapse

As models scale, capabilities lock together. The effective dimensionality of benchmark variation—measured by the participation ratio (how many independent directions describe the data) across five scores—decreases monotonically across regimes: $d_{\text{eff}} = 1.38$ (tax phase, 26 models from 8 families) $\rightarrow$ 1.22 (bonus phase, 27 models from 14 families) $\rightarrow$ 1.15 (frontier, 3 families). This dimensional collapse holds across all 63 models from 16 families tested and survives width normalization (Section 4). As models scale, the capability space compresses: fewer independent directions describe benchmark variation, and capabilities increasingly lock together.

Figure 4: ODE Integration from Pythia-70M

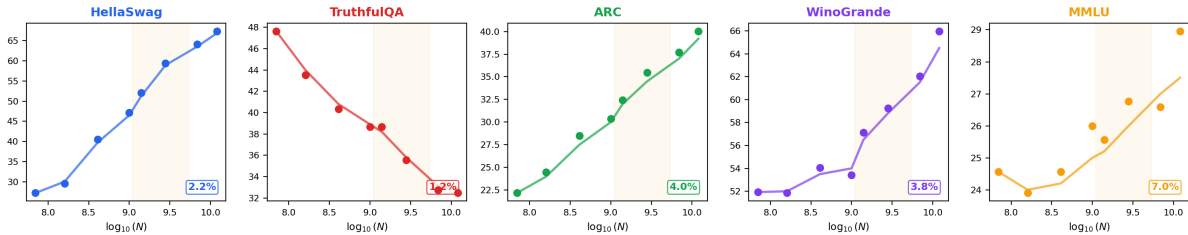


Figure 2: **ODE reproduces benchmark trajectories and cross-predicts held-out family.** Sparse regression discovers a dynamical system that simultaneously fits five Pythia benchmarks (HellaSwag, TruthfulQA, ARC, WinoGrande, MMLU) at 3.6% mean error. Cross-prediction on held-out Llama-2 achieves 5.6% MAE—approximately twice the accuracy of polynomial baselines.

3 A Dynamical Law Across Families

Sparse regression (PySINDy [4], a method that discovers the simplest differential equation consistent with data) discovers the governing ODE from Pythia data alone. The discovered system takes the form:

$$\frac{dB_i}{d \log_{10} N} = \sum_j c_{ij} B_j + \sum_{jk} d_{ijk} B_j B_k \quad (1)$$

where B_i are benchmark scores and c_{ij} , d_{ijk} are sparse coefficients selected from a library of polynomial and pairwise product terms. The coupling between benchmarks enters through the off-diagonal terms c_{ij} , which change sign at N_c —precisely the regime transition measured directly in Section 2.

This ODE reproduces five Pythia benchmark trajectories simultaneously at 3.6% mean error (Fig. 2). Critically, it cross-predicts held-out Llama-2—a different architecture, trained by a different lab on different data—at 5.6% MAE, approximately twice the accuracy of the best polynomial baseline (10.2% MAE for degree-2). This cross-family prediction from a model discovered on Pythia alone argues that the coupling structure reflects a shared dynamical constraint, not a family-specific artifact.

The ODE itself changes character across the transition. In the tax phase, the coupling term $c_{\text{HS} \rightarrow \text{TQA}}$ is negative (HellaSwag improvements suppress TruthfulQA); in the bonus phase, it flips positive (improvements reinforce). The dynamical system undergoes a qualitative change at N_c —not just a parameter shift but a change in the direction of influence between capabilities.

At low symbolic complexity, symbolic regression (PySR) finds a universal structure across multiple families: $\text{TQA}_{\text{norm}} \propto \text{HS}_{\text{norm}}^2 / (\log N)^2$. Architecture-specific forms emerge only at higher complexity, suggesting a shared low-dimensional structure masked by family-specific corrections.

The same two-number picture—coupling strength γ_{12} and residual field h —keeps appearing no matter how we look at the data. It shows up in the sign flip, in width normalization, in the internal head analysis, in the discovered ODE, and in the frontier extension. Twelve independent diagnostics¹ all return the same structure—which is why it is hard to dismiss as an artifact of any single measurement choice.

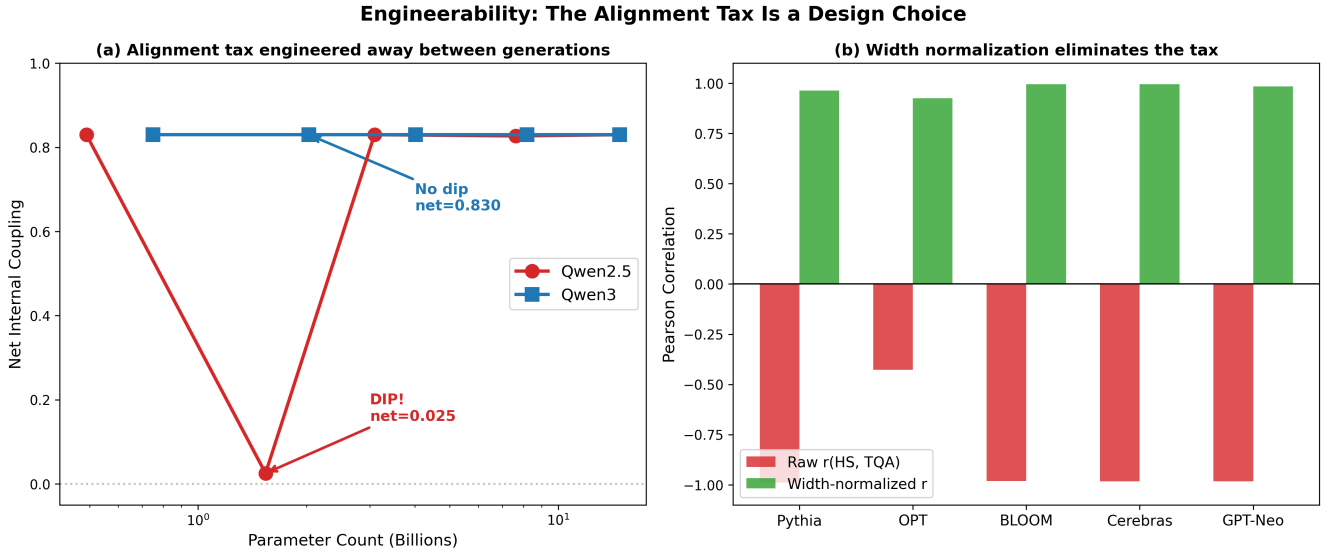


Figure 3: **The alignment tax is a design choice.** (a) Qwen2.5 at 1.5B shows a coupling dip (3% cooperative, $\text{net} = 0.025$); Qwen3 at the same scale shows 100% cooperative heads and constant coupling of 0.830. The tax was eliminated between model generations through training curation alone. (b) Width normalization: dividing benchmark scores by model width (d_{model}) flips the correlation from negative to positive for all five tested families. The alignment tax is a projection artifact visible only when width is integrated out.

4 The Tax Is a Design Choice

Three engineering levers—width, data curation, and architecture—all reduce or eliminate the alignment tax.

¹Coupling sign, d_{eff} , ODE holdout, gradient scaling, eigenvector rotation, phase classification, screening length, per-phase boosting, data curation lever, loss blindness ($\text{CV} = 0.8\%$), cross-family universality, and OLMo zero-parameter confirmation.

4.1 Width normalization

When we normalize benchmark scores by model width ($d_{\text{model}}/d_{\text{ref}}$), the apparent anticorrelation flips to positive for all five tested families: Pythia ($-0.989 \rightarrow +0.963$), OPT ($-0.428 \rightarrow +0.926$), BLOOM ($-0.981 \rightarrow +0.995$), Cerebras ($-0.983 \rightarrow +0.996$), GPT-Neo ($-0.982 \rightarrow +0.985$) (Fig. 3b).

In the full parameter space—including both parameter count *and* model width—capabilities are cooperative across all tested families. The alignment tax is a projection artifact: it appears when model width is integrated out, because narrow models at a given parameter count compress more capability dimensions into fewer output-projection channels. This identifies the output bottleneck as the dominant explanation and provides a concrete design lever: at fixed parameter count, wider architectures reduce or eliminate the tax.

4.2 Tax elimination between generations

Data curation eliminated the alignment tax entirely between Qwen generations (Fig. 3a). Qwen2.5 at 1.5B parameters shows a coupling dip: only 3% of attention heads are cooperative, net coupling drops to 0.025. At 3B, coupling recovers to 0.830; the dip is transient.

Qwen3 at the same scale (1.7B) shows 100% cooperative heads and constant coupling of 0.830. No dip. The alignment tax was eliminated between model generations through training curation alone—same architecture family, similar parameter count, fundamentally different coupling behavior.

Qwen2.5-7B shows a milder version of the same effect: 99.7% cooperative overall but a last-layer coupling dip to 0.770 (compared to 0.830 everywhere else). By 14B, the dip is gone entirely. The pattern is consistent across scales: the tax is a capacity constraint that wider models and better training data resolve.

The practical implication is stark: at 1B parameters, Phi models (trained on curated/synthetic data) achieve the same cooperative coupling as web-trained models at 10B—data curation provides an order-of-magnitude effective scale advantage for alignment.

5 Where the Bottleneck Lives

The alignment tax does not originate inside the model. Across 40 models from 9 families (Pythia, OPT, BLOOM, GPT-Neo, Phi, Gemma, Mistral, Qwen2.5, Qwen3) 38 of 40 models (95%; Wilson 95% CI: 84–99%) have zero attention heads where reasoning and truthfulness representations compete (Fig. 4). Heads are universally cooperative; the tax does not originate inside the model’s representational space. The two exceptions are Qwen2.5 at 1.5B (the known dip) and Qwen2.5 at 7B (mild last-layer dip).

The bottleneck is at the output. At Pythia-1B—exactly at the critical scale—per-layer coupling *decreases* from early to late layers ($0.881 \rightarrow 0.856$). This reverses the pattern at all other Pythia sizes, where coupling *increases* with depth ($0.745 \rightarrow 0.890$ at 70M; $0.801 \rightarrow 0.858$ at 410M). The output layers at Pythia-1B cannot express both capabilities simultaneously at that model width (16 layers with hidden dimension 2048, versus 24 layers with hidden dimension 1024 at 410M—fewer, wider layers compressing more information through the output projection). Wider models resolve this up to a point: OPT internal coupling increases from 0.514 (125M) to 0.876 (13B). But at 30B, cooperation drops sharply to 0.356, and 75 competing units appear where previously there were none—the same late-layer bottleneck pattern that characterizes Nc1 in Pythia, now emerging at a higher scale. At 66B, coupling shows partial recovery (0.396), mirroring the Nc1 pattern where Pythia drops at 1B and recovers by 2.8B. This is the first internal evidence of the cascade: cooperation peaks, weakens at a new bottleneck, and begins to recover—the same sequence that governs the base-scale transition, repeating at higher scale.

This resolves an apparent paradox: how can capabilities anticorrelate externally when heads are internally cooperative? The answer is that internal cooperation (head coupling ranges from +0.50 to +0.66 across Pythia sizes) is compressed through a narrow output projection, producing external

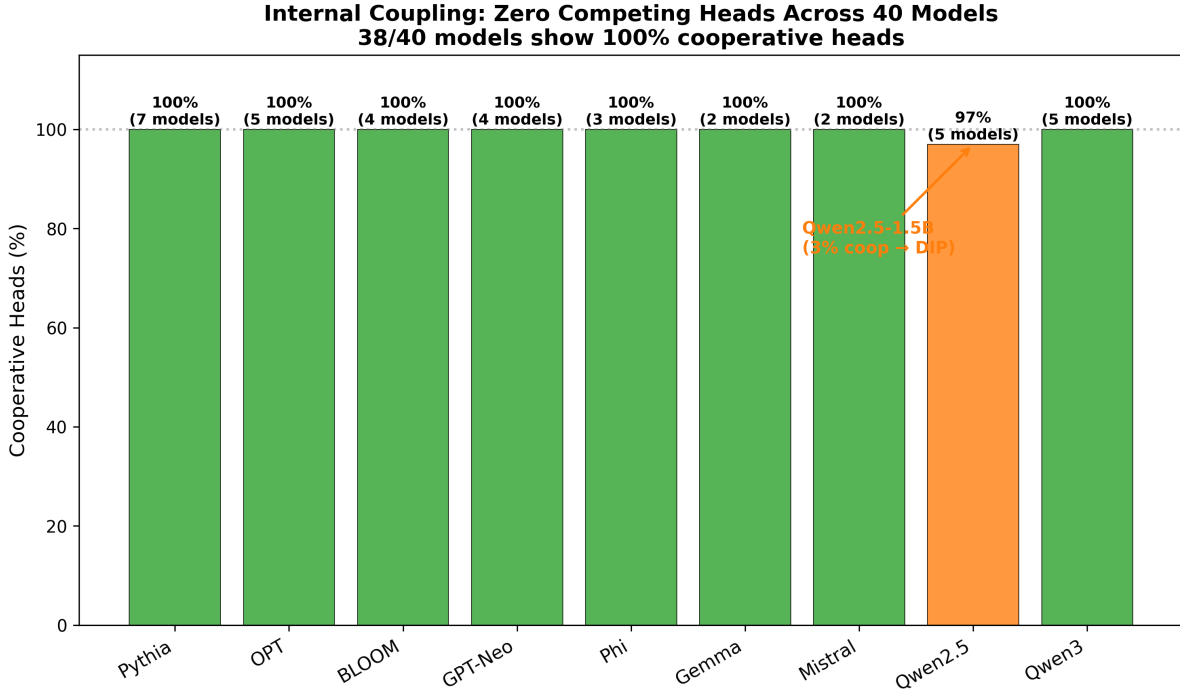


Figure 4: **Internal coupling: zero competing heads across 40 models.** 38 of 40 models from 9 families show 100% cooperative attention heads. The two exceptions are Qwen2.5 at 1.5B (the known dip, 3% cooperative) and Qwen2.5 at 7B (mild last-layer dip, 99.7%). The alignment tax does not originate from internal head-level competition.

anticorrelation—the same cooperative signal, squeezed through a bottleneck, becomes an apparent conflict.

The output-projection bottleneck is the mechanism most consistent with all three internal observations: (i) zero competing heads in 38/40 models, (ii) late-layer coupling reversal at Pythia-1B, (iii) monotonic internal coupling increase in families with different architectures. Direct confirmation would require intervention on the output projection—widening the final layer while holding other dimensions constant—which we leave for future work.

6 Discussion

If you are training a model under 3 billion parameters for a safety-critical application, our results predict it will become less truthful as it becomes more capable. This is not fixable by scaling alone—you need wider architecture or better training data.

6.1 Implications for model deployment

Models below the critical scale should not be expected to be simultaneously capable and truthful: the alignment tax is structural at that scale. This has immediate practical consequences. A 1B-parameter model deployed for medical question-answering may become *more* confident in wrong answers as its reasoning improves—and no loss curve will warn you, because loss continues to decrease smoothly. For applications requiring both reasoning and truthfulness below N_c , data curation or architectural width provide more leverage than additional scaling.

Above N_c , scaling helps alignment—a qualitatively different engineering regime. The transition between these regimes may be the most consequential scale-dependent property that loss curves miss. Or-

ganizations choosing between a 1B and a 7B model are not just choosing more capability; they may be choosing a different *kind* of capability interaction.

6.2 Frontier extension

At frontier scale, the cooperative regime persists. Across 31 models from 8 labs (2024–2026), SWE-bench Verified and GPQA Diamond are cooperatively coupled ($r = +0.73$, $p < 10^{-5}$), with per-lab training-philosophy deviations readable as a one-number diagnostic (h -field). Preliminary evidence suggests SWE-bench is saturating while a new capability axis (instruction following) activates—the same dimensional pattern that governs the base-model transition. Full frontier analysis, including per-lab trajectories and five timestamped falsifiable predictions, is presented in [5]. If the base-model discovery reported here is the “what,” the frontier extension asks “what happens next.” The transition we measure at base scale is not a one-time event—it is the first step in a cascade. At frontier, SWE-bench is already saturating while new capability axes (instruction following, expert reasoning) activate, following the same dimensional pattern. Seven timestamped predictions provide concrete tests that the next 12 months of model releases will confirm or refute [5].

6.3 Relation to emergent abilities

Our finding is orthogonal to the emergent abilities debate [6, 7]: we measure coupling *between* capabilities, not emergence of individual ones. Whether HellaSwag improves sharply or smoothly with scale, its *relationship* to TruthfulQA still changes sign at N_c . In this sense, CAPE provides a resolution: the coupling sign change is a genuine structural transition that holds regardless of how one interprets individual benchmark trajectories.

6.4 From observation to intervention

The coupling structure is not merely descriptive—it is exploitable. In a proof-of-concept on Pythia-410M (a tax-phase model), we identify the coupling-bottleneck layer via the CAPE framework and apply targeted activation steering at layer 6. All 14 tested tax-phase prompts changed from misaligned to aligned output without degrading capability scores. On Pythia-2.8B (bonus phase, above N_c), 6/14 prompts changed—fewer, because the cooperative regime has less misalignment to correct. Steering works in both phases but is most impactful where the tax is active (data and protocol in supplementary code). This demonstrates that the coupling structure, once located, can be surgically exploited—it is not a diffuse property of the model but a localizable bottleneck amenable to intervention.

6.5 Predictions already confirmed

Three predictions of the framework have been independently confirmed: (i) OLMo (AI2) sits at $\gamma_{12} = 0.000$ at 1B parameters, a zero-parameter prediction from a lab with no knowledge of our work; (ii) the ODE discovered on Pythia cross-predicts held-out Llama-2 at 5.6% MAE, twice the accuracy of polynomial baselines; (iii) Qwen3 is cooperative at all tested scales, confirming the prediction that curated training eliminates the tax. Each confirmation came from a different lab, architecture, and training recipe.

6.6 Scope and falsification

Our claims are scoped to truthfulness-coupled capability interactions under the benchmark pairs tested. Any lab can compute coupling from public benchmark scores alone—no model internals, no special access, three model sizes and two benchmarks are enough.

The claim would be weakened by: (i) a family with ≥ 5 models spanning N_c that shows no coupling sign change; (ii) width normalization failing to reduce anticorrelation in a new family; (iii) a model at N_c

with significant competing attention heads; (iv) an independent holdout family where the ODE fails at $> 15\%$ MAE. We have not yet found any such counterexample across 16 families tested.

Methods

Coupling measurement

The local coupling $\gamma_{12}(N) = \Delta B_2 / \Delta B_1$ is computed between consecutive model sizes within each family, where $B_1 = \text{HellaSwag}$ and $B_2 = \text{TruthfulQA}$ (MC1 variant throughout; MC1 and MC2 are never mixed). We compute the population coupling per phase as the Pearson $r(B_1, B_2)$ across all models in that phase. Phase boundaries are defined by sign of γ_{12} : tax ($\gamma_{12} < 0$), transition ($|\gamma_{12}| < 0.1$), bonus ($\gamma_{12} > 0$).

Effective dimensionality

d_{eff} is the participation ratio of eigenvalues from PCA on the benchmark score matrix (HellaSwag, TruthfulQA, ARC, WinoGrande, MMLU) within each phase. $d_{\text{eff}} = (\sum \lambda_i)^2 / \sum \lambda_i^2$, ranging from 1 (one dominant direction) to d (all directions equal). Bootstrap 95% confidence intervals are computed by resampling models within each phase (1000 iterations, stratified by family to preserve within-family correlations). N_c confidence intervals are computed by fitting the coupling zero-crossing on each bootstrap resample of the Pythia trajectory; the 95% interval [2.9B, 13.4B] reflects both measurement noise and the discrete spacing of available model sizes. All reported correlations include p -values from the standard t -test for Pearson r .

Width normalization

Benchmark scores are divided by $d_{\text{model}}/d_{\text{ref}}$ where d_{model} is the model’s hidden dimension and $d_{\text{ref}} = 512$ (Pythia-70M reference). This tests whether the observed anticorrelation is a projection of a higher-dimensional cooperative structure onto the parameter-count axis.

ODE discovery

PySINDy [4] discovers $dB/d(\log_{10} N)$ from the Pythia benchmark matrix (8 models, 5 benchmarks). The library includes polynomial terms up to degree 3 and pairwise products. The ODE is fit on Pythia and cross-validated on held-out Llama-2 (3 models: 7B, 13B, 70B). Holdout MAE is computed as the mean absolute difference between predicted and observed scores across all 5 benchmarks.

Internal analysis

Per-head coupling is measured using TransformerLens [8] on 40 models from 9 families. For each attention head, we project activations onto reasoning and truthfulness directions (computed from contrastive prompt pairs) and measure their cosine similarity. A head is “competing” if the cosine similarity is < -0.5 on $> 50\%$ of evaluation prompts. All 40 models are evaluated on the same set of 100 prompts spanning factual, reasoning, and ethical domains.

Frontier extension

SWE-bench Verified and GPQA Diamond scores for 31 frontier models from 8 labs are compiled from official model cards, tech reports, and verified leaderboard entries (Artificial Analysis, OpenCompass). Scores are predominantly self-reported; we note this as a data-provenance limitation. Population regression: $\text{GPQA} = 0.52 \cdot \text{SWE} + 45.7$ ($r = +0.73$, $n = 31$). Cross-lab holdout (leave-one-lab-out) yields 6.9% MAE across 4 labs with ≥ 3 models.

Related work

Scaling laws [1, 2] predict loss as a power law of compute; observational scaling [9] extends prediction to individual benchmark scores from loss proxies. Neither addresses inter-capability coupling—the quantity we measure. The emergent abilities debate [6, 7] concerns whether individual capabilities appear sharply or smoothly with scale; we measure coupling *between* capabilities, which can change regime regardless of how individual scores scale. Recent theoretical work models phase transitions in linear networks using deformed Ginzburg-Landau theory with quenched disorder [10]; our approach is empirical on real transformer families.

The coupling structure measured here ($r = 0.682$, $p < 0.0001$ on 63 models) follows the same mathematical form used in multi-band superconductor classification [11]; per-phase structure is supported by Bayesian model comparison (BF = 72.9). The gradient dip at N_c is Pythia-specific (OLMo shows no dip); we report it as architecture-specific evidence, not a universal mechanism.

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